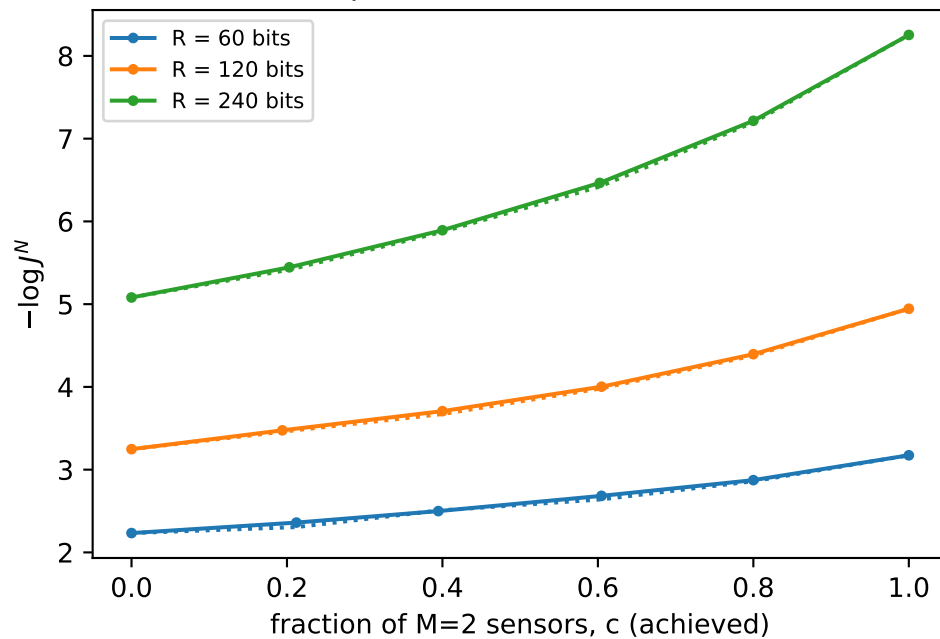
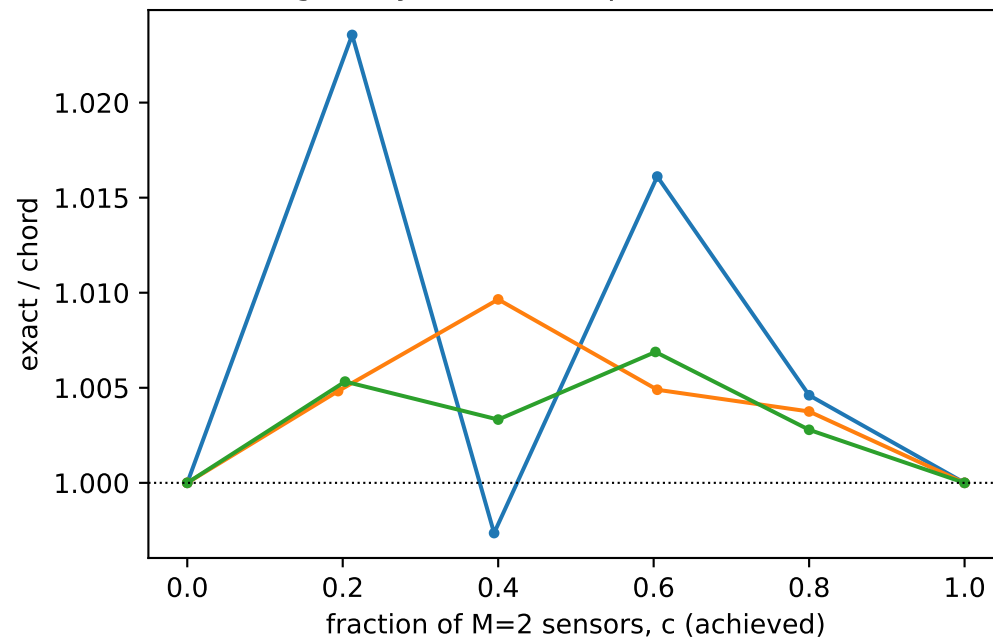


c-sweep M=2/M=4 at matched total rate, SNR -5 dB

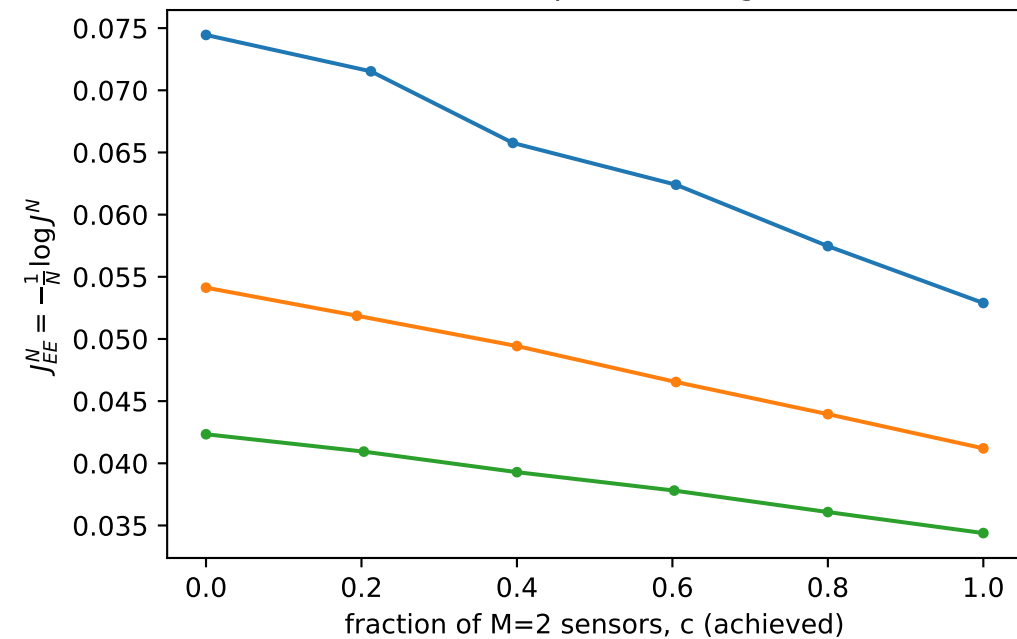
Total exponent vs c (dotted = chord)



Heterogeneity effect (1 = pure rate reallocation)



Normalized exponent (diagnostic)



Endpoints $c=0$ (all $M=4$) and $c=1$ (all $M=2$) are homogeneous, inside Assumption 1(iii); interior points are outside it. All banks at one rate target share the same total rate via $N = \text{round}(R/(2-c))$. Dotted chord = rate-weighted mix of the endpoint exponents; the exact/chord ratio in panel 2 isolates the heterogeneity effect from the $M=2$ vs $M=4$ bit-efficiency effect. Panel 3 divides by N , which varies with c .